

IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

Python Full Stack

Development With Project

AICTE – EduSkills Virtual Internship Program

Sujal Gautam

Roll No. 2410031651 | B.Tech CSE

June – August 2026

ABOUT THE STUDENT



**Sujal
Gautam**

B.Tech – Computer Science & Engineering

Candidate Profile

Roll Number

2410031651

Institute

IILM University, Greater Noida, U.P.

Programme

B.Tech CSE, Batch 2024–2028

Internship Domain

Python Full Stack Development With Project

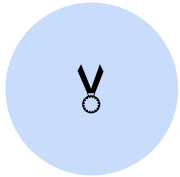
Duration

8 Weeks | June – August 2026

ABOUT THE PROGRAM

EduSkills, AICTE & Industry Learning

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme “Nation Building Through Skills.” It partners with the All India Council for Technical Education (AICTE) to run the AICTE–EduSkills Virtual Internship Program under the patronage of the National Internship Portal, Ministry of Education.



ISO Certified

ISO 9001:2015 & ISO/IEC 20000-1:2018 certified organization.



AICTE Patronage

Run under the National Internship Portal, Ministry of Education.



Full Stack Curriculum

Structured 8-week Python, FastAPI & React learning path.

PROBLEM STATEMENT

Building a Redis-Style In-Memory Cache

Students often understand caching concepts theoretically but rarely get hands-on practice building a working key-value store. This project, “In-Memory Cache with TTL Lite,” tackles that gap — implementing a simplified version of systems like Redis, supporting core operations, hit/miss tracking, time-based expiration, deletion, and statistics reporting, entirely in Python.

This project built the full workflow of an in-memory caching system:



What the Internship Set Out to Achieve



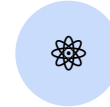
Python & Full Stack Basics

Build a strong foundation in Python, OOP, and full-stack development concepts.



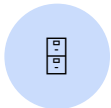
FastAPI & Validation

Build backend APIs using FastAPI with request handling and Pydantic validation.



React Fundamentals

Learn components, JSX, hooks (useState, useEffect) and dynamic UI building.



Database Integration

Integrate relational databases with SQLAlchemy ORM and perform CRUD operations.



In-Memory Cache with TTL

Design a key-value store with hit/miss tracking, TTL expiry, delete & reporting.



Capstone Deployment

Build and deploy a complete full-stack capstone application end to end.

The Toolkit Behind the Internship



Python

Core programming language



FastAPI

Backend API framework



Pydantic

Request & data validation



React

Frontend UI & hooks



SQLAlchemy

Database ORM & CRUD



Redux / Zustand

Frontend state management



Docker

Testing & deployment



TTL Caching

Key-value store, in-memory

From Command Input to Cache Report



The 8-Week Learning Journey

1 WEEK

Full Stack & Python Basics

Environment setup, data types, functions, OOP

2 WEEK

FastAPI & Data Validation

Backend APIs, routing, Pydantic validation

3 WEEK

React Fundamentals & Hooks

Components, JSX, useState, useEffect

4 WEEK

Frontend-Backend Integration

Reusable components, API calls, data rendering

5 WEEK

Database with SQLAlchemy

ORM, CRUD operations, relational models

6 WEEK

Advanced React & State Mgmt

React Router, Redux / Zustand for scale

7 WEEK

Testing & Deployment

Backend/frontend testing, Docker, cloud deploy

8 WEEK

Capstone Project

Full-stack e-commerce app: FastAPI + SQLAlchemy + React

EXPECTED OUTCOMES

Skills & Capabilities Gained

- ✓ Practical understanding of implementing in-memory data structures for key-value storage
- ✓ Experience managing data lifecycle through Time-To-Live (TTL) expiration mechanisms
- ✓ Skill in tracking and analyzing cache performance metrics like hits and misses
- ✓ Proficiency in Python for building stateful applications and structured JSON reports
- ✓ Hands-on exposure to FastAPI, React, and SQLAlchemy across the full stack
- ✓ Completed all learning modules and cleared the Final Assessment with an “Outstanding” grade

CERTIFICATION

Successfully Completed

Grade

O – Outstanding

Certificate ID

4d05a145fda54e73758d

Student ID

STU69e5258d296b31776625037

Duration

8 Weeks | June – August 2026





Thank You

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